Sequences — problem 9.41

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The sequence x_n is defined by

$$x_1 = 2, \qquad x_{n+1} = \frac{2 + x_n}{1 - 2x_n}, \qquad n = 1, 2, 3, \ldots .$$

Prove that (a) $x_n \neq 0$ for all n; (b) x_n is not periodic.